

Deep Reinforcement Learning for Adaptive Gain Tuning in Control of Teleoperation Manipulators with Joint Flexibility and Time-Varying Delays

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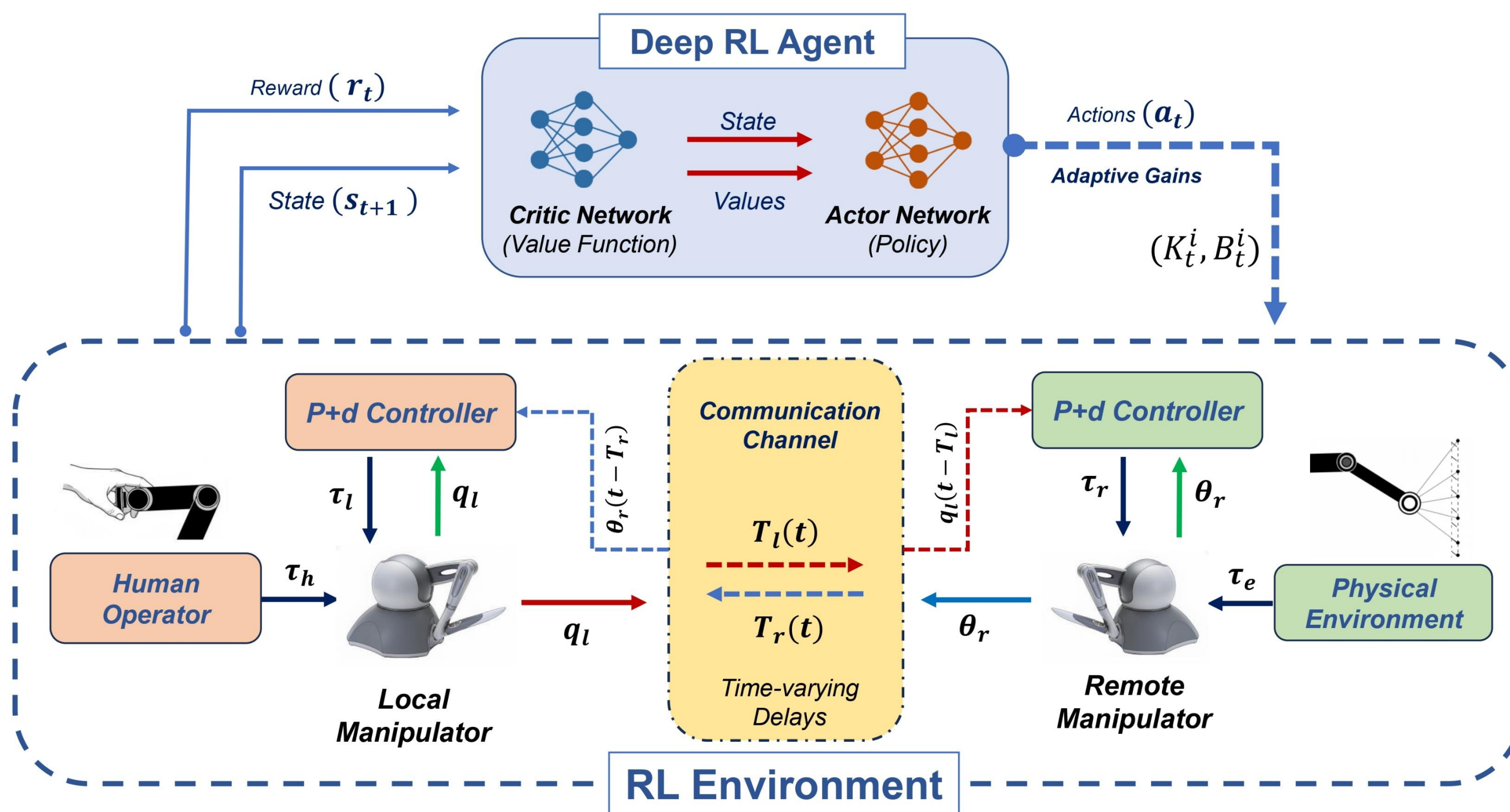


Fig. 1: Overview of the proposed control in bilateral teleoperation with joint flexibility and time-varying delays

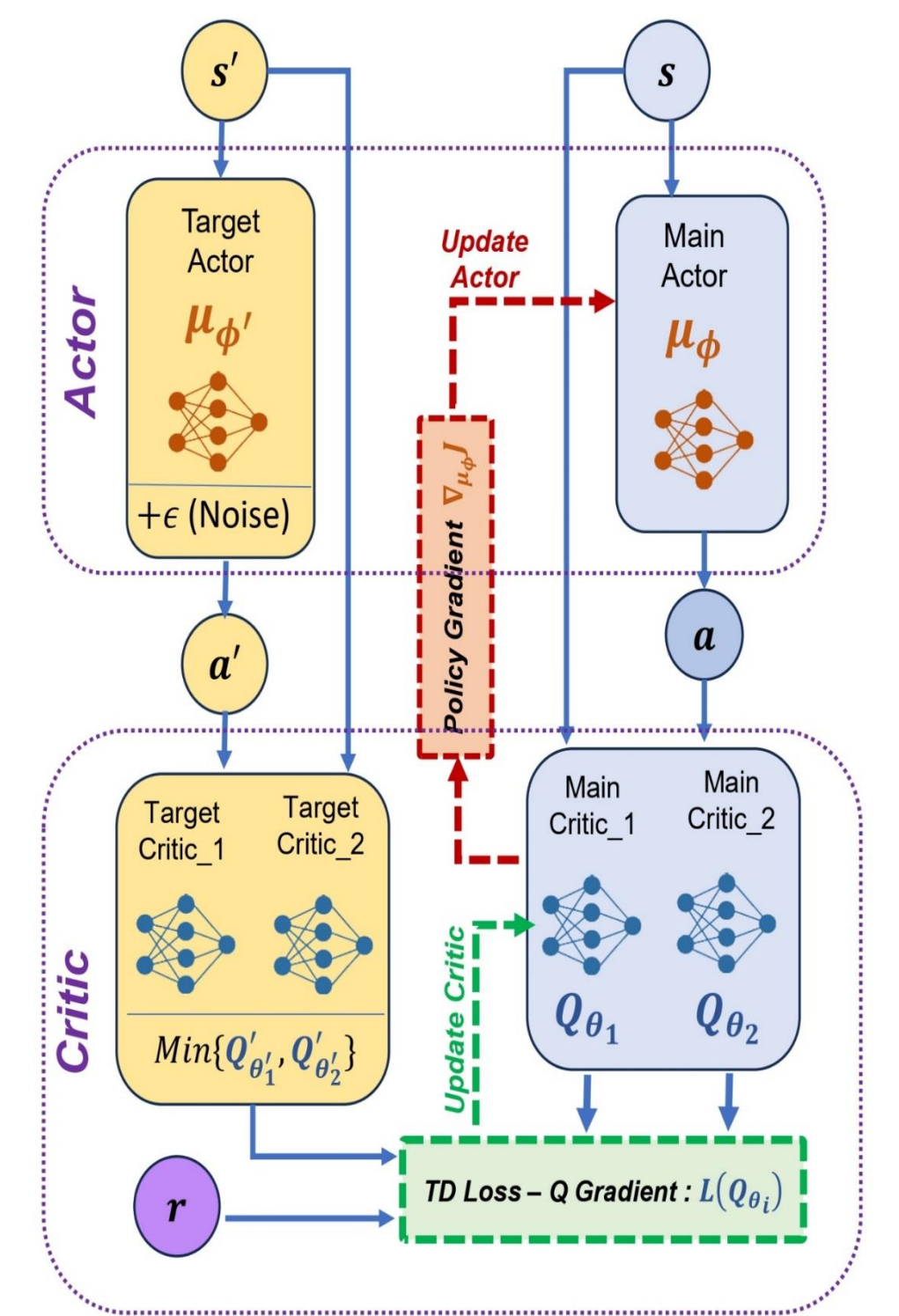


Fig2. TD3 network architecture

1. Introduction

Teleoperation enables humans to control remote robots in fields such as surgery, space, and rehabilitation. Major challenges include time delays, stability, transparency, and joint flexibility, which introduces vibrations and nonlinearities. Existing methods often handle delays *or* flexibility, but rarely both. Deep RL provides a promising way to tune control gains online. This motivates a hybrid DRL + P+d approach for flexible-joint teleoperation under time-varying delays.

2. Modeling of Flexible Joint Teleoperators

The system includes a rigid 2-DOF local manipulator and a flexible-joint remote manipulator. The remote side has motor positions θ_r and link positions q_r connected by stiffness S_r . Communication delays are time-varying and bounded. This model forms the basis for the proposed control design.

$$M_r(q_r)\ddot{q}_r + C_r(q_r, \dot{q}_r)\dot{q}_r + S_r[q_r - \theta_r] = -\tau_e$$

$$J_r\ddot{\theta}_r + S_r[\theta_r - q_r] = \tau_r$$

$$M_l(q_l)\ddot{q}_l + C_l(q_l, \dot{q}_l)\dot{q}_l = \tau_h - \tau_l$$

3. DRL-based Hybrid Control

A hybrid controller is used: a P+d controller guarantees baseline stability, while a TD3 agent adaptively tunes the remote proportional and damping gains. The RL agent observes tracking and damping errors and adjusts gains to reduce oscillations caused by joint flexibility and delays. Only the remote side is adaptively tuned.

$$\tau_l = K_l[q_l(t) - \theta_r(t - T_r(t))] + B_l\dot{q}_l(t)$$

$$\tau_r = -K_r[\theta_r(t) - q_l(t - T_l(t))] - B_r\dot{\theta}_r(t)$$

$$action(t) = [K_r^i(t), B_r^i(t)], \quad i = 1, 2.$$

4. Simulation and Results

Simulations use a 2-DOF flexible-joint slave and time-varying delays up to 1 s. The DRL-tuned gains stabilize the system and suppress vibrations.

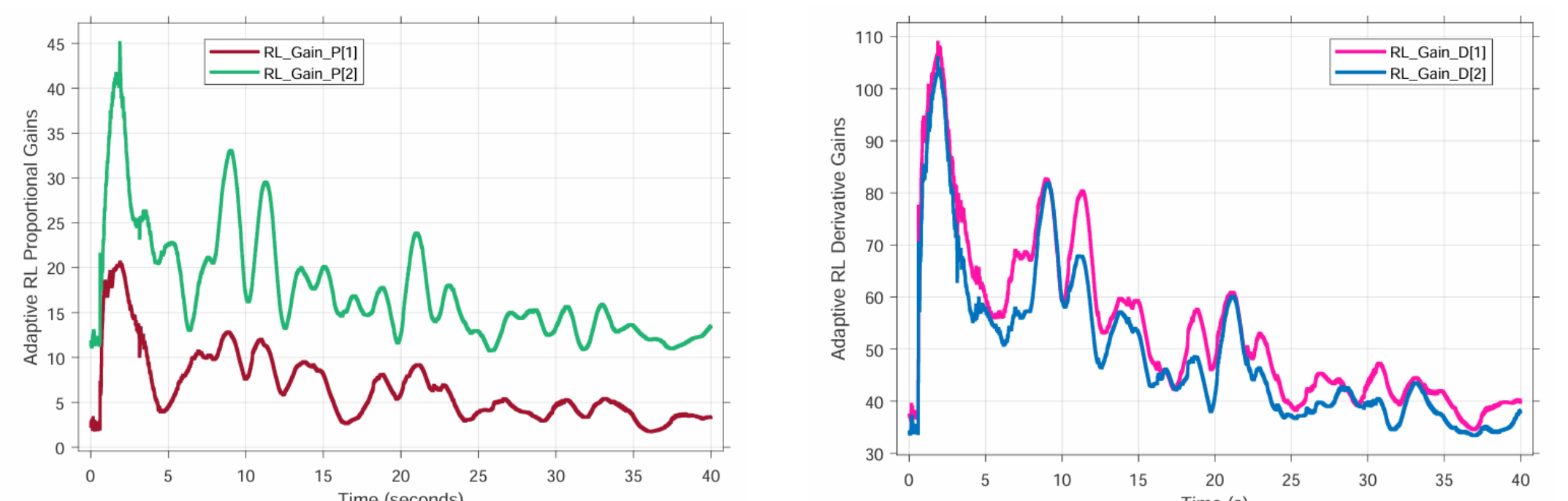


Fig 3. DRL-generated adaptive gains for $i=1,2$ (left) proportional gains K_t^i and (right) derivative gains B_t^i .